

# From Raw Opinions to Actionable Insights: User Perception Analysis Using Web Scraping, Social APIs, and Medallion Architecture

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## Introduction

The rapid growth of consumer-generated content offers valuable opportunities to monitor brand and product perception in real time. Yet converting unstructured, multi-source data into reliable intelligence demands a disciplined pipeline with clear separation of concerns.

This work presents an end-to-end system built on a Medallion Architecture (Bronze -> Silver -> Gold), integrating Twitter/X posts and technology news comments, progressively refined from raw ingestion through NLP enrichment to weekly analytical artifacts.

Sentiment is scored using VADER (-1.0 to +1.0) and mapped to three classes (positive / neutral / negative), with taxonomy-driven entity recognition linking each record to a predefined brand, product, and topic ontology.

The result is a fully automated intelligence layer that surfaces actionable weekly insights, including sentiment drift and anomaly detection, without manual annotation or model fine-tuning.

## Objective

To design and implement an automated, reproducible pipeline that collects, processes, and analyzes consumer opinions from heterogeneous digital sources, transforming unstructured text into structured weekly sentiment intelligence across technology brands, products, and topics through a layered Medallion Architecture.

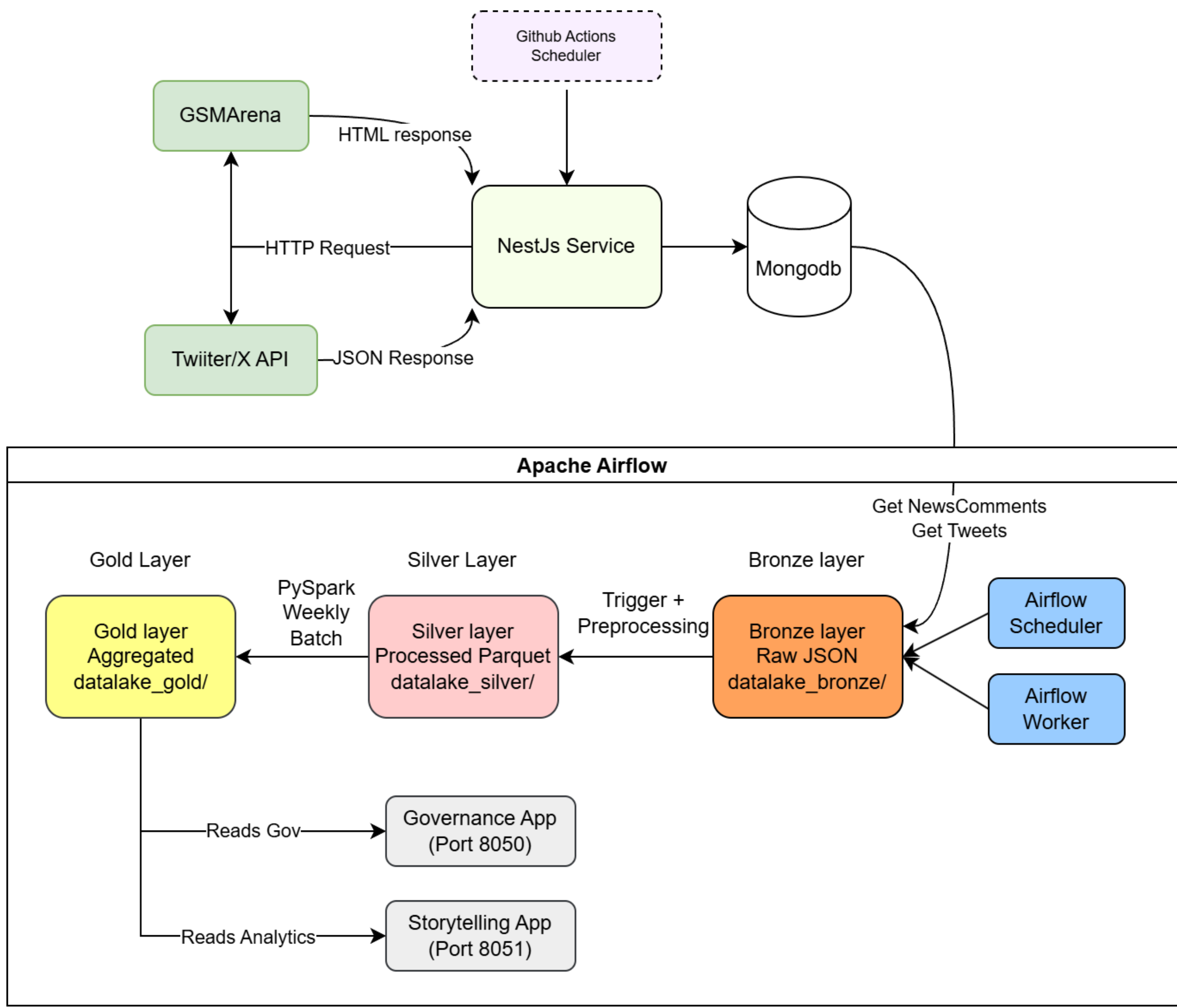
## Solution & Technical Architecture

The system follows a Medallion Architecture (Bronze -> Silver -> Gold) orchestrated by Apache Airflow. Since the pipeline runs on a local machine without guaranteed uptime, data collection was decoupled entirely: a NestJS service on Vercel handles GSMArena scraping and TwitterAPI.io calls, triggered on a fixed schedule by GitHub Actions. Raw data is persisted as JSON in MongoDB, forming the Bronze layer.

Airflow Silver DAGs consume these snapshots, applying flattening, deduplication, and NLP text cleaning, writing weekly-partitioned Parquet files via pandas and PyArrow, allowing incremental append within the same week.

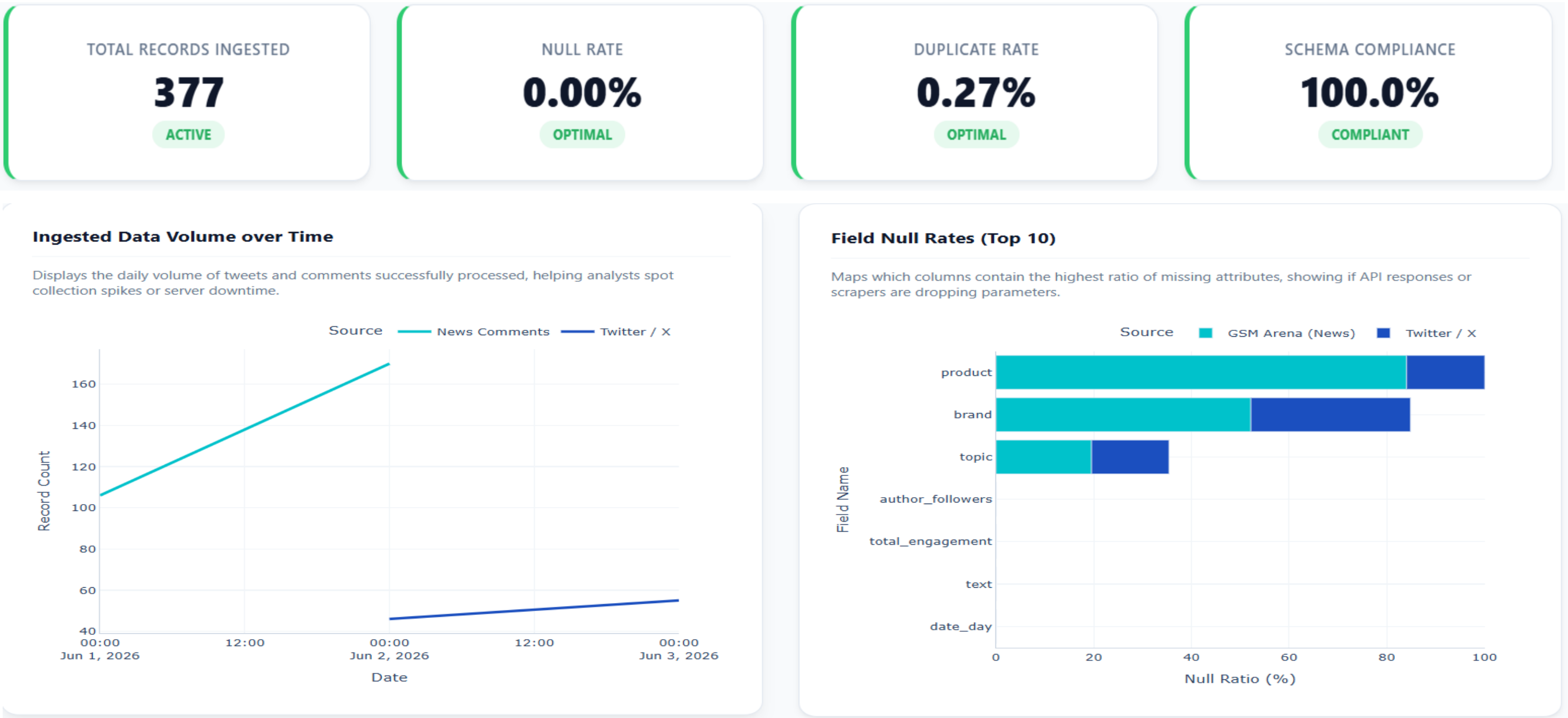
A weekly Gold DAG powered by Apache Spark and VADER classifies sentiment and matches records against a predefined brand, product, and topic taxonomy, producing eight analytical artifacts covering brand rankings, topic trends, product perception, and an executive storytelling summary.

Results are surfaced through two Dash / Plotly dashboards: a governance explorer focused on data quality monitoring, and a storytelling insights dashboard oriented toward business decision-making.



## Results

The pipeline successfully ingested and processed data from both sources with high quality metrics, including near-zero null and duplicate rates and full schema compliance, confirming the reliability of the decoupled collection architecture. The governance layer provided clear visibility into data volume, field completeness, and engagement outlier distribution across sources.



The storytelling dashboard translated raw records into actionable weekly intelligence, identifying dominant brands, discussion topics, and product sentiment trends. The automated alert system proved capable of detecting abrupt sentiment shifts in near real-time, surfacing critical signals that would otherwise require manual review of thousands of posts.

